

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/ Supplementary Winter Examination – 2024

Course: B. Tech

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTHM 505, Project Management

Max Marks: 60

Date: 22/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Completion of an activity on CPM network diagram, is generally known a. Event b. Node c. Connector d. All of the above	CO2	1
2	If the values of t_o , t_p of an activity are 4 and 15, what is the variance of the activity? a. 3.92 b. 3.22 c. 3.36 d. 3.87	CO3	1
3	The estimated time required to perform an activity is known a. Event b. Dummy c. Duration d. Float	CO2	1
4	The t_o , t_m & t_p for an activity are 4, 11, and 12 days resp. The T_E of activity is... a. 8 days CO2 c. 10 days d. 11 days	CO4	1
5	Critical Path a. Is always longest CO2 c. May be longest d. May be shortest	CO2	1
6	What does the direct cost of project include? a. Labour cost CO2 c. Equipment Cost d. All the above	CO4	1
7	Which of the following is correct sequence in calculating event time in CPM? a. EST-EFT-LST-LFT b. EST-LST-EFT-LFT c. EST-LFT-EFT-LST d. EST-EFT-LFT-EFT	CO2	1
8	Which of the following is a Bagged Material? a. Steel CO2 c. Brick d. Aggregates	CO1	1
9	CPM is used..... a. When duration for activities in known b. in Closing stage c. When the duration for activities in unknown d. for cost estimate	CO2	1

10	When the float of the activity is negative, that activity is called as.....				CO2	1
	a. Critical Activity	CO2	c. Super Critical Activity	d. Subcritical Activity		
11	Bar charts are suitable for....				CO1	1
	a. Minor Works	CO2	c. Large Projects	d. All of the above		
12	Critical path method				CO1	1
	a. Is an improvement upon bar chart method	b. Provides a realistic approach to daily problems	c. Avoids delays which are very common in bar charts	d. All the above		
Q. 2	Solve the following.				CO1	12
A)	Explain phases in project management.				CO2	6
B)	Differentiate between CPM and PERT					6
Q. 3	Solve the following.					12
A)	What is the necessity and data required for updating the network?				CO1	6
B)	Crash the network given above and find Project Duration (Days) and Project Cost.				CO3	6
	Activity	Normal Time	Crash Time	Normal Cost	Crash Cost	
	1 – 2	5	3	5000	10000	
	1 – 3	2	1	1500	3000	
	2 – 4	5	3	2000	3000	
	2 – 5	7	3	5000	8000	
	3 – 5	6	3	8000	10000	
	4 – 6	8	4	5000	6000	
	5 - 6	9	5	15000	18000	
Q. 4	Solve Any Two of the following.					12
A)	Explain briefly Direct cost and indirect cost for construction project.				CO4	6
B)	Draw the network and Determine the Project Duration.				CO3	6
	Activity	Optimistic Time	Most likely time	Pessimistic time		
	1 – 2	6	10	20		
	1 - 3	7	10	13		
	2 – 4	9	15	21		

		2 – 5	7	11	15							
		3 – 5	3	7	11							
		4 – 6	9	12	15							
		5 - 6	7	9	11							
	Find the Z Value if the Project mentioned above has to be completed in 22 days.											
C)	Discuss central limit theorem in detail.					CO4	6					
Q. 5	Solve Any Two of the following.						12					
A)	Explain Demand and Supply curve, Inflation and Cash flow diagram.					CO4	6					
B)	List the different types of interests. Explain any two.					CO5	6					
C)	Draw the network and calculate TE & TL for all activities for the following.					CO2	6					
	Activity (i-j)	1- 2	2-3	2- 4	3-5	3 – 6		4-5	4-7	5-8	6-8	7-8
	Duration (t ^{ij})	5	2	6	4	4		2	3	7	8	2
Q. 6	Solve Any Two of the following.						12					
A)	The Fixed Cost associated with a product is Rs. 50 Lakhs, the variable cost is Rs. 25, and the revenue from each item is Rs. 50. Calculate the Break-even quantity if the targeted profit is Rs. 10 Lakhs.					CO4	6					
B)	Explain the uses of computer software (Microsoft Project and Primavera) in project management.					CO5	6					
C)	Explain ISO 14000 in detail.					CO5	6					
	*** End ***											